Recall : definition of determinant

What is det A?

Suppose A is an $n \times n$ matrix.

- The determinant of A is a number which can be calculated by cofactor expansion across a row or down a column.
- The main property of det A is that A is invertible if and only if det $A \neq 0$.

Determinant of a 2×2 matrix

If $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ then

$$\det A = a_{11}a_{22} - a_{12}a_{21}$$

Sometimes we write this as $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$

Minors of an $n \times n$ matrix

Let A be an $n \times n$ matrix.

Suppose $1 \leqslant i \leqslant n$ and $1 \leqslant j \leqslant n$ are integers.

- The (i,j)-minor of A is the $(n-1) \times (n-1)$ matrix obtained by eliminating the i-th row and the j-th column of A. It is denoted by A_{ij}.

Example. $A = \begin{bmatrix} \mathbf{1} & \mathbf{-4} & \mathbf{5} \\ 0 & \mathbf{2} & \mathbf{1} \\ \mathbf{3} & \mathbf{2} & \mathbf{4} \end{bmatrix}$ then :

$$A_{31} = \begin{bmatrix} -4 & 5 \\ 2 & 1 \end{bmatrix}$$

$$A_{11} = \begin{bmatrix} 2 & 1 \\ 2 & 4 \end{bmatrix}$$

$$A_{22} = \begin{bmatrix} 1 & 5 \\ 3 & 4 \end{bmatrix}$$

$$A_{23} = \begin{bmatrix} 1 & -4 \\ 3 & 2 \end{bmatrix}$$

Determinant of a 3×3 matrix - row expansions

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \qquad \rightsquigarrow \qquad \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

- **Cofactor expansion across the first row :**

$$\det A = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

- **Cofactor expansion across the second row :**

$$\det A = -a_{21} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} + a_{22} \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} - a_{23} \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}$$

- **Cofactor expansion across the third row :**

$$\det A = a_{31} \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} - a_{32} \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} + a_{33} \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

One can also expand the determinant down any column :

Determinant of a 3×3 matrix - column expansions

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad \rightsquigarrow \quad \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

- Cofactor expansion down the first column :

$$\det A = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{21} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} + a_{31} \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix}$$

- Cofactor expansion across the second column :

$$\det A = -a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{22} \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} - a_{32} \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix}$$

- Cofactor expansion across the third column :

$$\det A = a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} - a_{23} \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} + a_{33} \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & 22 \end{vmatrix}$$

Determinant of an $n \times n$ matrix

Suppose $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$ is an $n \times n$ matrix.

- The determinant of A can be calculated by cofactor expansion across a row or down a column.
- The formula for the determinant of an $n \times n$ matrix depends on the formula for the determinant of an $(n-1) \times (n-1)$ matrix.

(i,j)-Cofactors

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

• Suppose A_{ij} denotes the (i,j)-minor of A. The (i,j)-cofactor of A is the number $C_{ij} = (-1)^{i+j} \det A_{ij}$.

The sign $(-1)^{i+j}$ is determined by the following checkerboard pattern of signs :

$$\begin{bmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \end{bmatrix}$$

E.g., $C_{11} = \det A_{11}$, $C_{12} = -\det A_{12}$, $C_{32} = -\det A_{32}$, etc.

Determinant of an $n \times n$ matrix

Suppose $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$ is an $n \times n$ matrix.

- **Cofactor expansion across the p-th row :**

$$\det A = a_{p1}C_{p1} + a_{p2}C_{p2} + \cdots + a_{pn}C_{pn}$$

- **Cofactor expansion across the q-th column :**

$$\det A = a_{1q}C_{1q} + a_{2q}C_{2q} + \cdots + a_{nq}C_{nq}$$

Example

Example

Calculate the determinant of $A = \begin{bmatrix} 3 & 0 & 1 & 2 \\ -2 & 1 & 0 & 2 \\ 1 & 3 & 4 & 2 \\ -3 & 2 & 1 & -1 \end{bmatrix}$ using expansion down the third column.

Solution. The formula is

$$\det A = a_{13}C_{13} + a_{23}C_{23} + a_{33}C_{33} + a_{43}C_{43} = 1 \cdot C_{13} + 0 \cdot C_{23} + 4 \cdot C_{33} + 1 \cdot C_{43}$$

Using the following checkerboard pattern of signs : $\begin{bmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{bmatrix}$

we can write the formula as :

$$\det A = 1\begin{vmatrix} -2 & 1 & 2 \\ 1 & 3 & 2 \\ -3 & 2 & -1 \end{vmatrix} - 0\begin{vmatrix} 3 & 0 & 2 \\ 1 & 3 & 2 \\ -3 & 2 & -1 \end{vmatrix} + 4\begin{vmatrix} 3 & 0 & 2 \\ -2 & 1 & 2 \\ -3 & 2 & -1 \end{vmatrix} - 1\begin{vmatrix} 3 & 0 & 2 \\ -2 & 1 & 2 \\ 1 & 3 & 2 \end{vmatrix}$$

Example (cont.)

$$\begin{aligned}\det A &= 1\begin{vmatrix}-2 & 1 & 2\\ 1 & 3 & 2\\ -3 & 2 & -1\end{vmatrix} - 0\begin{vmatrix}3 & 0 & 2\\ 1 & 3 & 2\\ -3 & 2 & -1\end{vmatrix} + 4\begin{vmatrix}3 & 0 & 2\\ -2 & 1 & 2\\ -3 & 2 & -1\end{vmatrix} - 1\begin{vmatrix}3 & 0 & 2\\ -2 & 1 & 2\\ 1 & 3 & 2\end{vmatrix}\\ &= 31 - 0 + 4(-17) + (-1)(-26) = -11\end{aligned}$$

Because we can calculate each of the 3×3 determinants. For example :

$$\begin{vmatrix}-2 & 1 & 2\\ 1 & 3 & 2\\ -3 & 2 & -1\end{vmatrix} = (-2)\begin{vmatrix}3 & 2\\ 2 & -1\end{vmatrix} - 1\begin{vmatrix}1 & 2\\ 2 & -1\end{vmatrix} + (-3)\begin{vmatrix}1 & 2\\ 3 & 2\end{vmatrix} = (-2)(-7) - (-5) + (-3)(-4) = 31$$

(The remaining ones are left as an exercise.)

Triangular matrices

- An $n \times n$ matrix is called upper triangular if every entry below the diagonal is zero.

$$\begin{bmatrix} -1 & 3 & -1 & 3 \\ 0 & 3 & 2 & 0 \\ 0 & 0 & -2 & 5 \\ 0 & 0 & 0 & -3 \end{bmatrix}$$

- An $n \times n$ matrix is called lower triangular if every entry above the diagonal is zero.

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -2 & 2 & 0 & 0 & 0 \\ 3 & 0 & -4 & 0 & 0 \\ 4 & 1 & 5 & 2 & 0 \\ 0 & -3 & 4 & 5 & -1 \end{bmatrix}$$

- A matrix is called triangular if it is either upper triangular or lower triangular.

Theorem

The determinant of a triangular matrix is equal to the product of its diagonal entries.

Example. Calculate the determinant of $A = \begin{bmatrix} -2 & 2 & 0 & 3 & 1 \\ 0 & 2 & -5 & 7 & -5 \\ 0 & 0 & 3 & 4 & -4 \\ 0 & 0 & 0 & 9 & -4 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}$

One can use row or column expansions, but it is easier to note that the matrix A is upper triangular, so $\det A = (-2)(2)(3)(9)(-1) = 108$

Row Operations and determinant

- If a multiple of one row of A is added to another row to produce a matrix B, then $\det A = \det B$.

Suppose $A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 0 \\ 0 & 1 & 4 \end{bmatrix}$. We calculate $\det A$ using cofactor expansion down the second column.

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

$$\begin{vmatrix} 1 & 3 & -2 \\ 2 & -1 & 0 \\ 0 & 1 & 4 \end{vmatrix} = -3\begin{vmatrix} 2 & 0 \\ 0 & 4 \end{vmatrix} + (-1)\begin{vmatrix} 1 & -2 \\ 0 & 4 \end{vmatrix} - 1\begin{vmatrix} 1 & -2 \\ 2 & 0 \end{vmatrix} = -24 - 4 - 4 = -32$$

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 0 \\ 0 & 1 & 4 \end{bmatrix} \xrightarrow{R_1 \to R_1 + R_2} \begin{bmatrix} 3 & 2 & -2 \\ 2 & -1 & 0 \\ 0 & 1 & 4 \end{bmatrix} = B$$

Then:

$$\begin{vmatrix} 3 & 2 & -2 \\ 2 & -1 & 0 \\ 0 & 1 & 4 \end{vmatrix} = -2\begin{vmatrix} 2 & 0 \\ 0 & 4 \end{vmatrix} + (-1)\begin{vmatrix} 3 & -2 \\ 0 & 4 \end{vmatrix} - 1\begin{vmatrix} 3 & -2 \\ 2 & 0 \end{vmatrix} = -16 - 12 - 4 = -32$$

Column operations and determinant

- If a multiple of one column of A is added to another column to produce a matrix B, then $\det A = \det B$.

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 0 \\ 0 & 1 & 4 \end{bmatrix} \qquad \det A = -32$$

Let B be obtained by adding (-2) times the first column to the third column.

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 0 \\ 0 & 1 & 4 \end{bmatrix} \xrightarrow{C_3 \to C_3 - 2C_1} \begin{bmatrix} 1 & 3 & -4 \\ 2 & -1 & -4 \\ 0 & 1 & 4 \end{bmatrix} = B$$

We can calculate $\det B$ by cofactor expansion across the third row.

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

$$\begin{vmatrix} 1 & 3 & -4 \\ 2 & -1 & -4 \\ 0 & 1 & 4 \end{vmatrix} = 0 \begin{vmatrix} 3 & -4 \\ -1 & -4 \end{vmatrix} - 1 \begin{vmatrix} 1 & -4 \\ 2 & -4 \end{vmatrix} + 4 \begin{vmatrix} 1 & 3 \\ 2 & -1 \end{vmatrix} = 0 - 4 - 28 = -32$$

Scaling a row or a column

- If a row (or a column) of a matrix is multiplied by a scalar c to produce B, then $\det B = c \det A$.

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 0 \\ 0 & 1 & 4 \end{bmatrix} \qquad \det A = -32$$

Suppose B is obtained from A by multiplying the second column by $-\frac{1}{2}$.

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 0 \\ 0 & 1 & 4 \end{bmatrix} \xrightarrow{C_2 \to -\frac{1}{2}C_2} \begin{bmatrix} 1 & -\frac{3}{2} & -2 \\ 2 & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & 4 \end{bmatrix} = B$$

We calculate $\det B$ by expansion down the second column.

$$\det B = \begin{vmatrix} 1 & -\frac{3}{2} & -2 \\ 2 & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & 4 \end{vmatrix} = -\left(-\tfrac{3}{2}\right)\begin{vmatrix} 2 & 0 \\ 0 & 4 \end{vmatrix} + \tfrac{1}{2}\begin{vmatrix} 1 & -2 \\ 0 & 4 \end{vmatrix} - \left(-\tfrac{1}{2}\right)\begin{vmatrix} 1 & -2 \\ 2 & 0 \end{vmatrix} = 12 + 2 + 2 = 16$$

Note that $\det B = -\frac{1}{2} \det A$.

Interchanging two rows or two columns

- If two rows (or two columns) of A are interchanged to produce B, then $\det A = -\det B$.

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 0 \\ 0 & 1 & 4 \end{bmatrix} \qquad \det A = -32$$

Suppose B is obtained from A by interchanging the first and the third column.

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 0 \\ 0 & 1 & 4 \end{bmatrix} \xrightarrow{C_1 \leftrightsquigarrow C_3} \begin{bmatrix} -2 & 3 & 1 \\ 0 & -1 & 2 \\ 4 & 1 & 0 \end{bmatrix} = B$$

We can calculate $\det B$ using cofactor expansion down the first column.

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

$$\det B = \begin{vmatrix} -2 & 3 & 1 \\ 0 & -1 & 2 \\ 4 & 1 & 0 \end{vmatrix} = -2 \begin{vmatrix} -1 & 2 \\ 1 & 0 \end{vmatrix} - 0 \begin{vmatrix} 3 & 1 \\ 1 & 0 \end{vmatrix} + 4 \begin{vmatrix} 3 & 1 \\ -1 & 2 \end{vmatrix} = 4 - 0 + 28 = 32$$

Properties of determinants

Theorem

Let A be an $n \times n$ matrix.

- $A \xrightarrow{R_i \to R_i + cR_j} B \rightsquigarrow \det B = \det A \quad$ (here $i \neq j$).
- $A \xrightarrow{C_i \to C_i + cC_j} B \rightsquigarrow \det B = \det A \quad$ (here $i \neq j$).
- $A \xrightarrow{R_i \to cR_i} B \rightsquigarrow \det B = c \det A.$
- $A \xrightarrow{C_i \to cC_i} B \rightsquigarrow \det B = c \det A.$
- $A \xrightarrow{R_i \leftrightsquigarrow R_j} B \rightsquigarrow \det B = -\det A \quad$ (here $i \neq j$).
- $A \xrightarrow{C_i \leftrightsquigarrow C_j} B \rightsquigarrow \det B = -\det A \quad$ (here $i \neq j$).

Example

Compute the determinant of $A = \begin{bmatrix} 1 & -3 & 5 & 4 & -7 \\ 1 & -3 & 5 & 4 & -7 \\ 0 & 5 & 0 & 2 & -2 \\ -4 & 3 & 2 & 1 & 3 \\ 1 & 1 & 2 & -2 & 6 \end{bmatrix}$

Solution The top two rows of A are the same. We have:

$$A = \begin{bmatrix} 1 & -3 & 5 & 4 & -7 \\ 1 & -3 & 5 & 4 & -7 \\ 0 & 5 & 0 & 2 & -2 \\ -4 & 3 & 2 & 1 & 3 \\ 1 & 1 & 2 & -2 & 6 \end{bmatrix} \xrightarrow{R_2 \to R_2 - R_1} \begin{bmatrix} 1 & -3 & 5 & 4 & -7 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 5 & 0 & 2 & -2 \\ -4 & 3 & 2 & 1 & 3 \\ 1 & 1 & 2 & -2 & 6 \end{bmatrix} = B \Rightarrow \det A = \det B$$

But $\det B = 0$ because we can expand it along the second row:

$$\begin{aligned} \det B &= b_{21}C_{21} + b_{22}C_{22} + b_{23}C_{23} + b_{24}C_{24} + b_{25}C_{25} \\ &= 0 \cdot C_{21} + 0 \cdot C_{22} + 0 \cdot C_{23} + 0 \cdot C_{24} + 0 \cdot C_{25} = 0 \end{aligned}$$

Therefore $\det A = 0$.

If a matrix has a zero row or a zero column, then its determinant is zero.

Example

$$\begin{vmatrix} 1 & 0 & 1 & 4 \\ -2 & 0 & 1 & -5 \\ \frac{1}{2} & 0 & 8 & -3 \\ -2 & 0 & -4 & 5 \end{vmatrix} = 0 \quad \text{and} \quad \begin{vmatrix} -3 & 4 & 1 & 12 \\ 2 & 0 & 1 & -5 \\ 0 & 0 & 0 & 0 \\ -2 & 6 & -4 & 5 \end{vmatrix} = 0$$

- If two rows of A are identical, then $\det A = 0$.
- If two columns of A are identical, then $\det A = 0$.

Example

$$\begin{vmatrix} -3 & 4 & 1 & 12 \\ 2 & 0 & 1 & -5 \\ -3 & 4 & 1 & 12 \\ -2 & 6 & -4 & 5 \end{vmatrix} = 0$$

Another method of computing $\det A$

Let A be an $n \times n$ matrix. Recall that we have three kinds of elementary operations :

1. Interchange : $R_i \leftrightsquigarrow R_j$.
2. Replacement : $R_i \to R_i + cR_j$.
3. Scaling : $R_i \to cR_i$.

- To calculate the E.F. of A, we only need Interchange and Replacement. The Scaling operation is only required when we want to obtain the R.E.F.
- Replacement does not change the determinant. Interchange changes the sign of the determinant.
- Suppose we row reduce A, using only interchanges and replacements, to obtain its echelon form :

$$A \xrightarrow{\text{Row Reduction}} B \qquad \textit{E.F.}$$

Then

$$\det A = (-1)^r \det B$$

where r is the number of interchanges done throughout the reduction process.

Example

Calculate det A where $A = \begin{bmatrix} 2 & -1 & 0 & 1 & 2 \\ 2 & -1 & -1 & 3 & 2 \\ 0 & 1 & 3 & 0 & 1 \\ 0 & 2 & 6 & 3 & 3 \\ 2 & 0 & 3 & 1 & 7 \end{bmatrix}$

We calculate the E.F. of the matrix :

$$\begin{vmatrix} 2 & -1 & 0 & 1 & 2 \\ 2 & -1 & -1 & 3 & 2 \\ 0 & 1 & 3 & 0 & 1 \\ 0 & 2 & 6 & 3 & 3 \\ 2 & 0 & 3 & 1 & 7 \end{vmatrix} \xrightarrow[R_5 \to R_5 - R_1]{R_2 \to R_2 - R_1} \begin{bmatrix} 2 & -1 & 0 & 1 & 2 \\ 0 & 0 & -1 & 2 & 0 \\ 0 & 1 & 3 & 0 & 1 \\ 0 & 2 & 6 & 3 & 3 \\ 0 & 1 & 3 & 0 & 5 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 2 & -1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 0 & 1 \\ 0 & 0 & -1 & 2 & 0 \\ 0 & 2 & 6 & 3 & 3 \\ 0 & 1 & 3 & 0 & 5 \end{bmatrix}$$

$$\xrightarrow[R_5 \to R_5 - R_2]{R_4 \to R_4 - 2R_2} \begin{bmatrix} 2 & -1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 0 & 1 \\ 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$ E.F.

The last matrix is upper triangular and its determinant is $(2)(1)(-1)(3)(4) = -24$.

Since there was only one interchange in the above reduction process, we have

$$\det A = (-1)^1 \cdot (-24) = 24$$

Example

Calculate $\det A$ where $A = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 1 & 2 & 5 & 1 \\ 0 & 3 & -1 & 3 \\ 2 & 1 & 0 & -1 \end{bmatrix}$.

Solution.

$$\begin{vmatrix} 1 & 3 & 1 & 2 \\ 1 & 2 & 5 & 1 \\ 0 & 3 & -1 & 3 \\ 2 & 1 & 0 & -1 \end{vmatrix} \xrightarrow[R_4 \to R_4 - 2R_1]{R_2 \to R_2 - R_1} \begin{vmatrix} 1 & 3 & 1 & 2 \\ 0 & -1 & 4 & -1 \\ 0 & 3 & -1 & 3 \\ 0 & -5 & -2 & -5 \end{vmatrix}$$

$$\xrightarrow[R_4 \to R_4 - 5R_2]{R_3 \to R_3 + 3R_2} \begin{vmatrix} 1 & 3 & 1 & 2 \\ 0 & -1 & 4 & -1 \\ 0 & 0 & 11 & 0 \\ 0 & 0 & -22 & 0 \end{vmatrix} \xrightarrow{R_4 \to R_4 + 2R_3} \begin{vmatrix} 1 & 3 & 1 & 2 \\ 0 & -1 & 4 & -1 \\ 0 & 0 & 11 & 0 \\ 0 & 0 & 0 & 0 \end{vmatrix} = B \quad \text{E.F.}$$

We didn't use any interchanges. Therefore $\det A = \det B = 0$.

Example

Suppose that $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = -5$. What is $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} - 2a_{11} & a_{22} - 2a_{12} & a_{23} - 2a_{13} \\ \frac{1}{2}a_{31} & \frac{1}{2}a_{32} & \frac{1}{2}a_{33} \end{vmatrix}$?

Solution.

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} - 2a_{11} & a_{22} - 2a_{12} & a_{23} - 2a_{13} \\ \frac{1}{2}a_{31} & \frac{1}{2}a_{32} & \frac{1}{2}a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ \frac{1}{2}a_{31} & \frac{1}{2}a_{32} & \frac{1}{2}a_{33} \end{vmatrix} \quad (R_2 \to R_2 + 2R_1)$$

$$= (\frac{1}{2}) \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \quad (R_3 \to 2R_3)$$

$$= -\frac{5}{2}.$$

Theorem

A square matrix A is invertible if and only if $\det A \neq 0$.

Question. Let A be an $n \times n$ matrix. Suppose c is a scalar. How are $\det A$ and $\det cA$ related?

Solution. $\det(cA) = c^n \det A$.

Example

Suppose $A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 0 \\ 0 & 1 & 4 \end{bmatrix}$.

We calculated $\det A$ in the beginning of the lecture :$\det A = -32$.

Let us now calculate $\det\big((-2)A\big)$

$$\begin{vmatrix} -2 & -6 & 4 \\ -4 & 2 & 0 \\ 0 & -2 & -8 \end{vmatrix} = (-2)\begin{vmatrix} 1 & -6 & 4 \\ 2 & 2 & 0 \\ 0 & -2 & -8 \end{vmatrix} = (-2)^2\begin{vmatrix} 1 & 3 & 4 \\ 2 & -1 & 0 \\ 0 & 1 & -8 \end{vmatrix} = (-2)^3\begin{vmatrix} 1 & 3 & -2 \\ 2 & -1 & 0 \\ 0 & 1 & 4 \end{vmatrix}$$

$$= (-8)\cdot(-32) = 256$$

Theorem

For a square matrix A we have $\det A^T = \det A$.

Theorem

If A and B are $n \times n$ matrices, then

$$\det(AB) = (\det A)(\det B)$$

Example. Let $A = \begin{bmatrix} 1 & 2 \\ -2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 1 \\ -5 & 1 \end{bmatrix}$.

Then :

$$\det A = 7 \quad \text{and} \quad \det B = 3$$

Also :

$$AB = \begin{bmatrix} -12 & 3 \\ -11 & 1 \end{bmatrix} \quad \Longrightarrow \quad \det(AB) = 21.$$

Theorem

Suppose A and B are $n \times n$ matrices.

- If A and B are invertible then AB is also invertible.

  Because $\det A \neq 0$ and $\det B \neq 0$, so $\det(AB) = (\det A)(\det B) \neq 0$.

- If at least one of A and B is not invertible, then AB is not invertible either.

  Because $\det(AB) = (\det A)(\det B) = 0$.

- In particular, $\det(A^k) = (\det A)^k$.

  If A is invertible, then $\det(A^{-1}) = (\det A)^{-1} = \dfrac{1}{\det A}$